INCH-POUND

A-A-59380A
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SUPERSEDING
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3 December 1998

COMMERCIAL ITEM DESCRIPTION

TENSIOMETER, WIRE ROPE

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) outlines the minimum requirements for three types of hand-held wire rope (cable) tensiometers. The tensiometers will be used to measure static tension in aircraft control cables. The cables range from 1/16 inch to 5/16 inch diameter. These commercially available tensiometers measure tension up to 200, 450, and 1600 pounds within ±5 percent of the applied load. Only those products that meet or exceed the requirements outlined in this CID are acceptable.

2. CLASSIFICATION. The tensiometers are classified by tension ranges as follows:

Type I	300 to 1600 pounds
Type II	10 to 200 pounds, direct reading
Type III	150 to 450 pounds, direct reading

3. SALIENT CHARACTERISTICS.

3.1 General. The tensiometer shall be easy to operate and read. It shall clamp quickly and securely onto cables. The tensiometer shall be capable of being operated using one hand with bulky gloves. It shall have a zero adjustment setting. The tension reading shall be capable of being locked, so that the tension can be read after the tensiometer is removed from the cable.

Beneficial comments, recommendations addition deletions, clarifications, etc., and any data which may improve this document should be sent to: Warner-Robins ALC, WR-ALC/TILCC, 420 Second Street, Suite 100, Robins AFB, GA 31098-1640.

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3.2 Type I. The Type I tensiometer may include a calibration chart outlining the readings for each increment of load and each cable diameter. The chart, if used, shall be dated, certified, and attached to the back of the tensiometer or to the inside of the carrying case. It shall be protected by transparent material.

3.2.1 Adapters. Interchangeable cable adapters, if used with Type I tensiometers, shall be securely attached to the tensiometer during use. Each adapter shall be marked to indicate which cable diameters it is used with.

3.2.2 Type I dimensions and weight. When the tensiometer is in the storage configuration, the dimensions shall be no greater than 6.5 inches long, 4.5 inches wide, and 1.5 inches thick. Weight shall be no greater than 3 pounds.

3.3 Types II and III. Type II and Type III tensiometer may include a calibration chart outlining the readings for each increment of load and each cable diameter. The chart, if used, shall be dated, certified, and attached to the back of the tensiometer or on the inside of the carrying case. It shall be protected by transparent material.

3.3.1 Direct Reading Dial. The type II and III tensiometers shall have a direct reading dial calibrated in pounds of tension. A means shall be provided for locking the dial needle at any position produced by the tension load on the cable. This will permit the removal of the tensiometer from the cable for reading.

3.3.2 Cable diameter gage. The tensiometer shall include an integral gage that indicates the diameter of the cables.

3.3.3 Test bar. A test bar shall be provided to check the calibration of the tensiometer. When the tensiometer is set to the smallest cable diameter and 5 readings are averaged, the tensiometer shall indicate the load marked on the bar within ±2 percent of the value marked. The tensiometer's serial number shall be marked on the bar.

3.3.4 Type II and III dimensions and weight. When the tensiometer is in storage configuration the dimensions shall be no greater than 9.5 inches long, 6 inches wide, and 2 inches thick. The weight shall be no greater than 3 pounds.

3.4 Cable characteristics. The tensiometer shall indicate static tension in cable construction types 7 X 7 and 7 X 19 found in MIL-DTL-83420, "Wire Rope, Flexible, for Aircraft Control" Type I (non-jacketed) and Table I below. The tensiometer shall not be damaged by cable tensions that exceed specified ranges.

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TABLE I. Cable characteristics and tensiometer accuracy.

Type	Tension Range (lbs.)	Cable Diameter (Inches)								Accuracy ± (percent of applied load
		1/16	3/32	1/8	5/32	3/16	7/32	¼	5/16	
I	300 to 550					X	X	X	X	5%
	550 to 1600					X	X	X	X	5 %
II	10 to 200	X	X	X						5%
	30 to 200				X	X				5%
	80 to 200						X	X		5%
III	150 to 450			X	X	X	X	X		5%

3.5 Accuracy. The tensiometer shall indicate the tension in the cables listed in Table I above within the accuracy specified in Table I. The differences between successive check tests at any cable tension shall not exceed 3 percent.

3.6 Calibration. The calibration method shall be determined by the manufacturer, provided that their results can be verified by the calibration procedures utilized by the services.

3.7 Environmental requirements. The tensiometer shall operate and indicate accurately in environments between minus 20 and plus 120 degrees Fahrenheit with relative humidity between 10 and 95 percent.

3.8 Marking. Each tensiometer shall be marked with the manufacturer's name, the model number, serial number, National Stock Number (NSN), cable diameter (in inches), tension range (in pounds), and accuracy. The marking shall be legible and permanent.

3.9 Operating Instructions. Operating instructions shall be provided with each tensiometer.

3.10 Carrying case. The tensiometer shall be provided with a durable carrying case that shall protect the tensiometer and all accessories from damage during use, storage, and transport.

4. REGULATORY PROVISIONS.

In accordance with Section 23.403 of the Federal Acquisition Regulation (FAR), the Federal Government's policy is to acquire items composed of the highest percentage of recoverable materials practicable consistent with maintaining a satisfactory level of competition and without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from recovered materials.

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5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of the CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The offered tensiometer must have been on the commercial market for at least 2 years. At least 50 tensiometers must have been sold to commercial (non-military) customers.

5.3 Product sample(s). Product samples are required for examination and testing to determine whether the offered tensiometer meets the requirements of this CID, provides accurate and repeatable measurements, and has acceptable form, fit, function, and suitability for use.

6. PACKAGING.

6.1 Preservation, packing, or marking requirements shall be as specified in the contract or order.

7. NOTES.

7.1 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified in conversion tables contained in the latest version of FED-STD-376 and all other requirements of this CID are met. If the manufacturer elects to use metric dimensions as the basis upon which to derive or base a mechanical feature, a request shall be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

7.2 Source of documents. Federal and Military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

7.3 Ordering data. The following information should be provided at the time of order:

a. Title, CID document number, and date of document.
b. Type of tensiometer required (see Section 2).
c. Number of product samples required (see 5.3).
d. Packaging requirements (see 6.1).

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7.4 Cross-reference. The following is a cross-reference between the classification types in MIL-T-38760 and this CID:

MIL-T-38760 Type	A-A-59380 Type	Range (lbs.)
C7	I	300 to 1600
C8	II	10 to 200
C9	III	150 to 450

7.5 Part or identifying number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor:

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Type: 1 - Type I 300 to 1600 pounds
2 - Type II 10 to 200 pounds
3 - Type III 150 to 450 pounds

CID Document Number

7.6 Possible acceptable products. The following commercial products may meet the salient characteristics of this CID:

TABLE III. Possible acceptable products.

Type	Manufacturer	Model No.	National Stock Number
I	Opti Manufacturing Corp.	T5-8005-110-000	6635-00-530-1130
II	Sigmation, Inc. Opti Manufacturing Corp.	19738 (Type C8) T60 1001C8-1A	6635-01-455-6940 6635-00-530-1128
III	Sigmation, Inc. Opti Manufacturing Corp.	19739 (Type C9) T60 1001C9-1A	6635-01-375-2019 6635-00-530-1129

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MILITARY INTERESTS:
Custodians:
Army – GL1
Navy – AS

REVIEWER:
DLA – GS

CIVIL AGENCY
COORDINATING ACTIVITY:
GSA – FSS

Preparing Activity:
Air Force – 84

Agent:
Air Force – 99

Project Number: 6635-0201

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